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PTE 311 — Past Questions & Materials
by Renance

Course: PTE 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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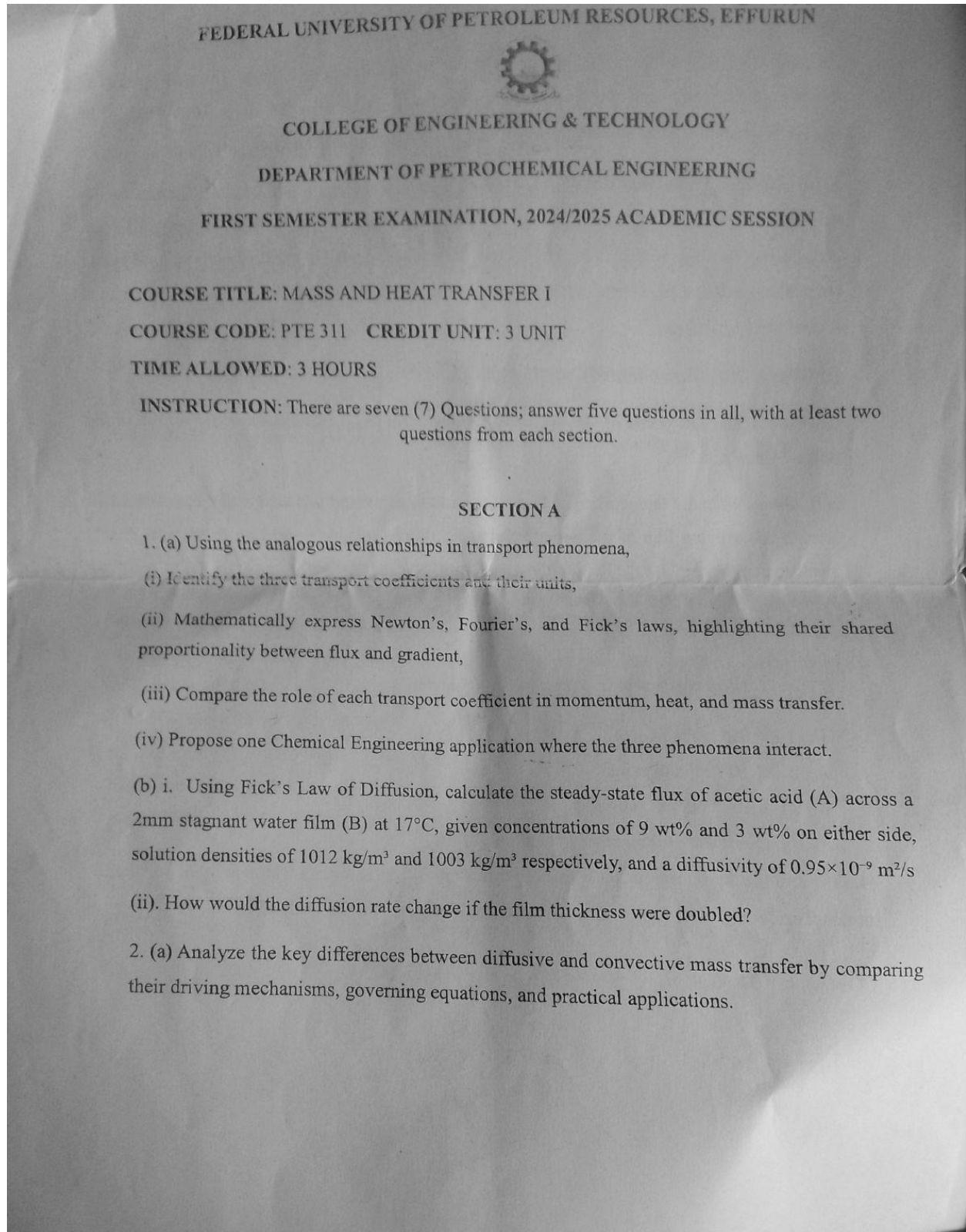
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Paper 1 of 1 — PTE 311 - 2024/2025 - First Semester - MASS AND HEAT TRANSFER I

Level 200 • First Semester • PastQuestion

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(b) Air at 20 °C ($\rho = 1.205 \text{ kg/m}^3$, $\nu = 15.06 \times 10^{-6} \text{ m}^2/\text{s}$, $D = 4.166 \times 10^{-5} \text{ m}^2/\text{s}$) flows over a tray (length = 320mm, width = 420 mm), full of water with a velocity of 2.8m/s. The total pressure of moving air is 1 atm, and the partial pressure of water present in the air is 0.0068 bar. If the temperature on the water surface is 150 °C, calculate the evaporation rate of water.

3. (a) LifeSaver Pharmaceutical Plc. is considering a Polyethylene film 0.04 cm thickness for use in packaging its newly formulated drug (Tracunizole) at 30 °C. The partial pressure of oxygen outside the package is 0.21 atm, and inside the package is 0.015 atm. Assuming steady state conditions, and that the resistance to diffusion outside and inside the film is negligible compared to the film's resistance.

(i) What is the diffusion flux of Oxygen ($\text{kgmol/m}^2\text{s}$) through a m^2 cross-section

(ii). How many kilograms of Oxygen pass per day through a square meter cross-section of the polyethylene film?

(iii). Given the same temperature and pressure conditions, would you advise the pharmaceutical firm to use a nylon film of the same thickness as a more suitable packaging for its new drug? Justify your answer. Use the Permeability data from Table 1 (attached)

Hint: P_m (Permeability of polyethylene) = $4.17 \times 10^{-12} \frac{(\text{m}^3 \text{O}_2 \text{ (stp)})}{\text{s.m}^2 \text{ atm/m}}$

(b). The diffusion coefficient of oxygen is approximately:

- $1.8 \times 10^{-5} \text{ m}^2/\text{s}$ in air
- $2.1 \times 10^{-9} \text{ m}^2/\text{s}$ in water
- $3.0 \times 10^{-14} \text{ m}^2/\text{s}$ in rubber

Explain these differences in magnitude.

4. (a) From the data provided, calculate the diffusion of CO_2 in air at a temperature of 25 °C and atmospheric pressure using the Chapman-Enskog Equation for Gas Diffusivity

CO_2 (A): $\sigma = 3.996 \text{ \AA}$; $\frac{\epsilon}{K} = 190\text{K}$, $M_A = 44$

Air (B): $\sigma = 3.167 \text{ \AA}$; $\frac{\epsilon}{K} = 97\text{K}$, $M_B = 29$

The table below may be used to estimate the collision integral

kT/ϵ	2.00	2.10	2.20	2.30
Ω	1.075	1.057	1.041	1.026

- (b) (i). Differentiate between the three modes of heat transfer
- (ii). As a process engineer at PowerGen Utilities, you must evaluate a 45 mm-thick copper heat exchanger plate ($k=370 \text{ W/m}\cdot^\circ\text{C}$) with face temperatures of 350°C (exhaust side) and 50°C (coolant side). Calculate the steady-state heat flux and total heat transfer for a 12m^2 plate

SECTION B

1. A pump draws 261.6 L/s of a liquid solution, with a density of 3.57 slug/ft^3 , from an open storage feed tank of large cross-sectional area through a 3.068-in ID suction. The pump discharges its flow through a 0.17225ft – ID line to an open overhead tank. The end of the discharge line is 50ft above the level of the liquid in the feed tank. The Friction losses in the piping system is $10\text{ft} - 16\text{f/1b mass}$. What pressure must the pump develop, and what horsepower must the pump have if its efficiency is 65% ? The flow is turbulent.
 - a. Water is flowing at a rate of 500.2 gpm through a horizontal nozzle. The nozzle is attached at the upstream, and the frictional forces are considered negligible. The upstream ID is 0.0635m , resultant force on the nozzle. The density of the water is 1000kg/m^3
2. Oil flows through a 1000-foot cast-iron pipe with a 10-inch diameter at a rate of 157.73 L/s
 Given that: Oil density: 1.65 slug/ft^3 , Kinetic Viscosity: 1.20 ft , Pump efficiency: 0.88

Determine

(ai) The Reynolds number

(aii). What is the power to be supplied by a pump to the fluid?

(aiii) Fluid the Brake horsepower of the pump

(b) A fluid with a density of 1000kg/m^3 and a dynamic viscosity of $0.001\text{ Pa}\cdot\text{s}$; flows through a pipe of diameter $D = 0.05\text{m}$ at an average velocity of $V = 2\text{m/s}$. Determine the Reynolds number.

3.(a) Water at 20°C flows through a 11811-inch -long steel pipe with a diameter of 7.874in and absolute roughness of 0.0001476ft . The head loss is known to be 204.7245in . Find the volumetric flow rate of water in the pipe

(b). Water with a density of 998kg/m^3 is flowing at a steady mass flow rate through a uniform diameter pipe. The entrance pressure of the fluid is $68.9\text{ KN/m}^2\text{ a/s}$ in the pipe, which connects to a pump that supplies 155.4J/kg of fluid flowing in the pipe. The exit section of the pipe is 3.05m higher than the entrance, and the exit pressure is $137.8\text{KN/m}^2\text{ a/s}$. The Reynolds number in the pipe is above 4000 in the system. Determine the frictional loss in the pipe system.

List of Formulas

(i)

$$N_A = \frac{m_A}{A} = D_{AB} \left[\frac{C_{A1} - C_{A2}}{x_2 - x_1} \right]$$

(ii)

$$D_{AB} = 0.0043 \frac{T^{3/2}}{p [V_A^{1/3} - V_B^{1/3}]^2} \left[\frac{1}{M_A} + \frac{1}{M_B} \right]^{1/2}$$

(iii)

$$\frac{D_1}{D_2} = \left(\frac{p_2}{p_1} \right) \left(\frac{T_1}{T_2} \right)^{3/2}$$

(iv)

$$D = \frac{1}{\left(\frac{x_B}{D_{AB}} \right) + \left(\frac{x_C}{D_{AC}} \right) + \left(\frac{x_D}{D_{AD}} \right)}$$

(v)

$$D_{AB} = 0.001858 \frac{T^{3/2}}{p (\sigma_{AB})^2 \Omega} \left[\frac{1}{M_A} + \frac{1}{M_B} \right]^{1/2}$$

where,

D_{AB} = Mass diffusivity of gas species B diffusing through another gas species A, cm²/s.

T = Absolute temperature, K.

p = Total pressure in atmospheres = $(p_A + p_B)$.

σ_{AB} = Collision diameter in Å (Angstroms),

Ω = Collision integral, and

M_A, M_B = Molecular weights of gas species A and B respectively.

(vi)

$$\sigma_{AB} = \frac{\sigma_A + \sigma_B}{2}$$

Table 1: Diffusivities and Permeability in Solids

Solute (A)	Solid (B)	T (K)	D_{AB} , Diffusion Coefficient [m^2/s]	Solubility, S [$m^3 \text{ solute (STP)} / m^3 \text{ solid} \cdot \text{atm}$]	Permeability, P_{AB} [$m^3 \text{ solute (STP)} / s \cdot m^2 \cdot \text{atm}/m$]
H ₂	Vulcanized rubber	298	$0.85(10^{-9})$	0.040	$0.342(10^{-10})$
O ₂		298	$0.21(10^{-9})$	0.070	$0.152(10^{-10})$
N ₂		298	$0.15(10^{-9})$	0.035	$0.054(10^{-10})$
CO ₂		298	$0.11(10^{-9})$	0.90	$1.01(10^{-10})$
H ₂	Vulcanized neoprene	290	$0.103(10^{-9})$	0.051	
		300	$0.180(10^{-9})$	0.053	$6.53(10^{-12})$
H ₂	Polyethylene	298			$4.17(10^{-12})$
O ₂		303			$1.52(10^{-12})$
N ₂		303			$0.029(10^{-12})$
O ₂	Nylon	303			$0.0152(10^{-12})$
N ₂		303			$0.15-0.68 \times 10^{-4}$
Air	English leather	298			$0.16(10^{-10})$
H ₂ O	Wax	306			$0.91-1.82(10^{-10})$
H ₂ O	Cellophane	293			$4.65(10^{-12})$
He	Pyrex glass	292			$7.0(10^{-15})$
H ₂	SiC ₂	293	$2.5(10^{-14})$	0.01	
H ₂	Fe	293	$2.59(10^{-13})$		
Al	Cu	293	$1.3(10^{-14})$		

Additional information (conversion factors)

$$1 \text{ btu} = 778.17 \text{ ft} \cdot \text{lb}_f = 1055.06 \text{ J}$$

$$1 \text{ hp} = 550 \text{ ft} \cdot \text{lb}_f/\text{s} = 0.7457 \text{ kW}$$

$$1 \text{ ft} \cdot \text{lb}_f/\text{lb}_m = 2.9890 \text{ J/kg}$$

$$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$$